

8. Project Demonstration

8.1. Model In-Action Walkthrough

The trained machine learning model is used to predict whether a customer's credit card application will be approved or rejected based on their financial and personal information.

Step 1: User Input Collection

The system collects customer details such as age, income, employment status, credit history, education level, and other relevant attributes required for prediction.

Step 2: Data Preprocessing

The collected input data is cleaned and transformed into a suitable format. Missing values are handled, categorical values are encoded, and numerical features are scaled according to the preprocessing steps applied during model training.

Step 3: Feature Processing

The processed data is converted into feature values that match the format expected by the trained machine learning model.

Step 4: Model Prediction

The input features are passed to the trained classification model (such as Random Forest, Logistic Regression, or other selected algorithms). The model analyzes the patterns learned from historical credit card application data.

Step 5: Prediction Output

The model generates a prediction result:

Approved – if the applicant meets the learned approval criteria.

Rejected – if the applicant does not satisfy the required conditions.

8.2. Advantages

-Faster Decision Making

The system quickly predicts credit card approval status, reducing the time required for manual application evaluation.

-Improved Accuracy

Machine learning models analyze historical data patterns and provide more consistent predictions.

-Reduces Human Errors

Automated prediction minimizes mistakes caused by manual processing and subjective decisions.

-Efficient Risk Assessment

The model helps identify potential risks by analyzing customer financial information and credit behavior.

8.3. Limitations & Future Scope

-Feature Sparsity: The current classifier relies on four metrics. Future iterations will include categorical variables (e.g. `'Education_Type'`, `'Housing_Type'`) in the preprocessing script.

-Dependence on Data Quality: The accuracy of the prediction depends on the quality, completeness, and reliability of the training data.

- Limited Prediction Capability:The model may not consider all real-world factors that influence credit card approval decisions.
- Bias in Training Data:If the training dataset contains biased patterns, the model may produce unfair predictions.